Tezpur University

Spring Semester End Examination-2025

Course Code: IT517 **Course Title: Pattern Recognition**

Full marks=50 *Time= 2 hours*

(The figures in the right-hand margin indicate full marks for the questions)

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1. Select the correct answers from the options given for each question: **1×5=5**
   a. Which among the following clustering methods is most sensitive to outliers?
      i. k-means
      ii. DBSCAN
      iii. k-medoids
      iv. Ward's method
   b. The point of a fuzzy set where its membership value is 0.5 is called
      i. support
      ii. core
      iii. crossover point
      iv. normality
   c. The number of layers necessary to classify samples into two classes separated by a hyperplane is
      i. Five
      ii. two
      iii. three
      iv. four
   d. Feature Selection based on probabilistic separability-based criterion function tries to
      i. maximize between class distances
      ii. minimize within class distances
      iii. maximize separation between density functions
      iv. maximize both between class and within class distances
   e. The minimum distance between a sample in one cluster and a sample in the other cluster is taken to be the distance between the two clusters in
      i. single-linkage algorithm
      ii. complete-linkage algorithm
      iii. average-linkage algorithm
      iv. ward's method

2. State True or False **1×4=4**
   a. In branch and bound Feature Selection method, feature subsets generation is complete.
   b. Perceptron algorithm cannot classify samples which are not linearly separable.
   c. PCA find projection where inter-class distances are maximum.
   d. k-medoids algorithm more time efficient than k-means algorithm.

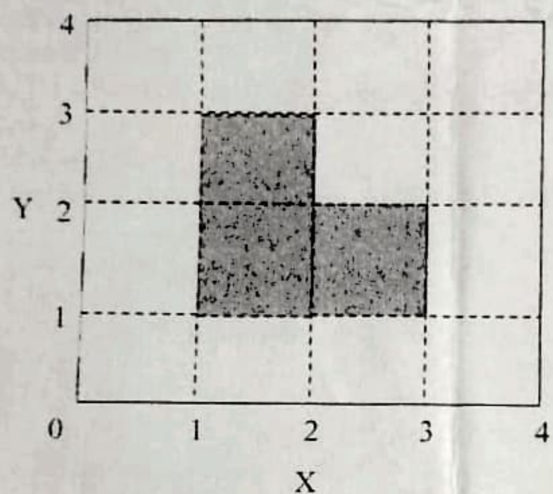
3. Answer the following question in short (any four): **3×4=12**
   a. Explain how Linear Discriminant Analysis (LDA) is different from Principal Component analysis.
   b. Discus the limitations of k-means clustering algorithm.
   c. Explain how elbow method is used to estimate the number of clusters.
   d. Explain why the output clusters of the DBSCAN algorithm are disjoint.

e. Explain how probabilistic separability criterion functions work.

4. (a) Construct a search tree of the Brach and Bound algorithm for selection of 2 attributes from 6. 8

OR

(b) Construct a neural net using 0/1 discontinuous threshold function that outputs 1 if the pattern is in the shaded region as shown in the following figure and 0 otherwise. 8



5. Perform a hierarchical clustering of the data given below using single-linkage algorithm and Manhattan distance. Show the distance matrices and the dendrogram. 8

Samples	1	2	3	4	5	6	7
X	0	0.5	1.0	2.0	2.0	6.0	7.0
Y	0.5	1.0	2.0	1.5	8.0	3.0	3.0

OR

Describe the K-Medoids algorithm. Discuss the comparative advantages/disadvantages of this algorithm with K-Means algorithm 6+2

6. Describe BCubed precision and recall metrics of cluster quality measure. 6

7. Derive the updating rules for the cluster centroids and membership values of Fuzzy k-means algorithm. 7

Tezpur University
Spring Semester Mid Examination-2025

Course Code: IT517 **Course Title: Pattern Recognition**

Full marks=30 *Time= 90 minutes*

(The figures in the right-hand margin indicate full marks for the questions)

1. Select the correct answers from the options given for each question: 1×4=4
   a. Which among the following clustering methods can find clusters of arbitrary shapes?
      i. k-means
      ii. DBSCAN
      iii. k-medoids
      iv. Agglomerative Hierarchical
   b. *Whitening Transform* converts a multivariate normal distribution into
      i. A univariate distribution.
      ii. A distribution having a diagonal covariance matrix.
      iii. A distribution having a covariance matrix equal to the identity matrix.
      iv. A distribution having a covariance matrix proportional to the identity matrix.
   c. Which of the following is not an essential criteria of cluster quality measure
      i. Cluster homogeneity
      ii. Cluster completeness
      iii. Smaller number of clusters
      iv. Rag bag
   d. The number of layers necessary to classify samples into two decision regions using neural network where one region is convex and the other is complement of it is
      i. five
      ii. two
      iii. three
      iv. four

2. Answer the following question in short (**any three**) : 3×4=12
   a. A biased coin shows heads 60 percent of the time on average. What is the probability that two of five flips will show heads?
   b. Explain why k-means clustering algorithm is more sensitive to noise than k-medoids.
   c. Explain how cross validation can be used to estimate the number of clusters.
   d. Explain why the output clusters of the DBSCAN algorithm are disjoint.

3. A series of eight samples gave the following results:

Class	Feature x	Feature y
A	0	1
A	1	0
A	0	1

Class	x	y
A	0	0
A	1	1
A	0	
B	0	1
B	1	0
B	1	0
C	0	1
C	1	1

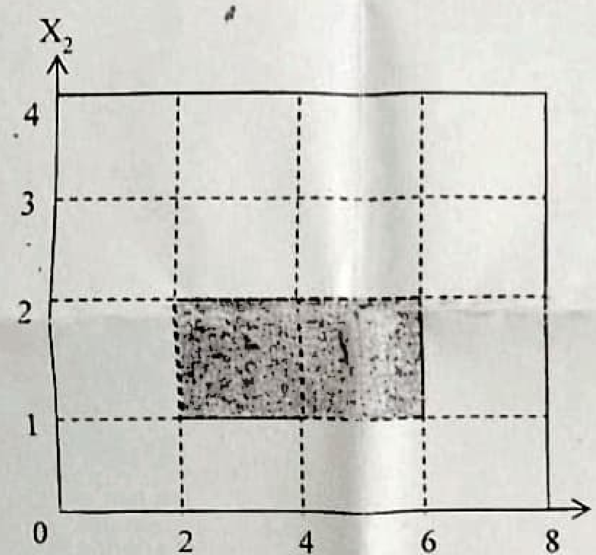
$P(A / x=0, y=1)$

Find out the probability that an unknown sample with $x = 0$ and $y = 1$ come from class A using Bayesian decision theory. 4

4. (a) Describe the sequential minimum squared error (MSE) algorithm for multiple outputs. 5

OR

(b) Construct a neural net using 0/1, discontinuous threshold function that outputs 1 if the pattern is in the shaded region as shown in the following figure and 0 otherwise. 5



5. Find out the maximum likelihood estimate for the probability of obtaining Head while tossing a coin. 5

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Test - I, Spring 2025

Department of Computer Science and Engineering

IT 517 Pattern Recognition

Time: 30 minutes | Total Marks: 10

$4 \times 0.5 = 2$

1. Answer the following objective questions.

   a. In Bayesian decision theory, for zero-one loss function the likelihood ratio $P(x|w_1)/P(x|w_2)$ is equal to .... (Fill in the blank)

   b. A transformation which converts an arbitrary multivariate normal distribution into a spherical one is called .... transformation. (Fill in the blank)

   c. If $P(A/B) = P(A)$ the the two events A and B are independent the. (True/False)

   d. In a two category classification, a loss function which penalizes error of classifying class 2 patterns as class 1 pattern more than the converse results in reducing of the decision region of class 2. (True/False)

2. (a) What is the probability that a classifier with 90% accuracy will correctly classify 9 objects out of 10? 4

   OR

   (b) Show that the normal density is symmetric around the mean.

3. Show that the discriminant functions of Bayesian classifier where the covariance matrices of classes are equal an proportionate to the identity matrix in linear 4
